

Image Browsing for Efficient Image Annotation

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Abstract—Image annotation is a work intensive and difficult task. Approaches for automatic image annotation are not sufficiently mature yet to be used in practise. In this paper, we show how an image database visualisation and browsing tool can be used for efficient and effective annotation of image databases. In contrast to other approaches, which typically present images in a linear fashion, we employ a visualisation where images are arranged so that visually similar images are located close to each other. Organisation on a grid layout prevents image overlap and hence contributes to a clear presentation. Large image databases are handled through a hierarchical storage structure where each image in the visualisation can correspond to a cluster of images that can be expanded by the user. Experimental results indicate that annotation can be performed faster on our proposed system.

Keywords—Image databases, image browsing, image annotation, hue sphere.

I. INTRODUCTION

Information is becoming more and more visual. Consequently, methods to effectively deal with this information, in particular with images, are highly sought after. Unfortunately, textual annotations are not available for most images, while annotating images is a challenging task. Manual annotation is work intensive and difficult, while automatic annotation techniques [1] are not sufficiently mature yet to be used in practise. In this paper we show how an image database visualisation and browsing tool can be used for efficient and effective annotation of image databases. In contrast to other approaches, which typically present images in a linear fashion, we employ a visualisation where images are arranged so that visually similar images are located close to each other. Organisation on a grid layout prevents image overlap and hence contributes to a clear presentation. Large image databases are handled through a hierarchical storage structure where each image in the visualisation can correspond to a cluster of images that can be expanded by the user. Experimental results show that annotation can be performed faster on our proposed system.

II. IMAGE DATABASE BROWSING

Browsing systems provide an overview of the image database to allow for intuitive navigation through the image collection [2], [3], [4], [5]. This is particularly the case when images are arranged according to mutual similarity [6]. That is, visually similar images are located close to each other in the browsing interface. One of the main challenges of such image database navigation systems is the limited screen size. Consequently, images are typically displayed as thumbnails arranged in such a way that the user can intuitively understand

what kind of images are contained in the dataset [2]. The way these thumbnails are arranged is often based on mapping of the feature space (typically using dimensionality reduction techniques such as multi-dimensional scaling [7]), or based on clustering concepts [8]. Once an image set is visualised, effective tools are needed for the user to be able to interact with the dataset in an efficient and intuitive way [3].

The visualisation approach we employ in this paper is based on the hue sphere browsing system presented in [9], [10], [11] and further developed into an on-line application [12] to provide an accessible, efficient and effective approach to image database browsing. The reason for adopting a spherical visualisation space is that users will be familiar with the concept of a globe and how to navigate on one. Each image is described by its median colour. However, rather than employing the standard RGB colour space, the HSV space is used [13] which humans find more intuitive. Of this, only the hue H and value (brightness) V attributes are taken, since the saturation descriptor is deemed less important for characterising image content. H and V are then transformed to longitude and latitude co-ordinates which describe the location of the image thumbnail on the sphere. The sphere can be rotated to focus on images with different colours or tilted to look at brighter or darker images.

Since the co-ordinates are extracted directly from the images, this approach is very fast and much more efficient than methods such as multi-dimensional scaling which rely on a comparably slow and computationally expensive iterative convergence procedure. At the same time, it does not compromise the achieved image database visualisation, since, as has been shown [14], the average image colour is at least as effective a descriptor for image browsing as high-dimensional feature vectors (e.g. based on colour distributions). In addition, the features employed are intrinsically suitable for query-based image retrieval as well [15], which allows for a combination of browsing and querying.

As overlapping and occlusion of image thumbnails will have a negative impact on the browsing experience [6], images are placed on a regular grid structure so that they do not overlap each other. Clearly, and in particular for larger image sets, this will mean that more than one image can be mapped to the same cell. This is handled through a hierarchical approach of storing and accessing the image collection and is performed based directly on the grid mapping, leading to a hierarchical tree based on clustering images. However, in contrast to clustering-based visualisation methods such as [8] there is no need for computationally expensive clustering algorithms. Rather, the

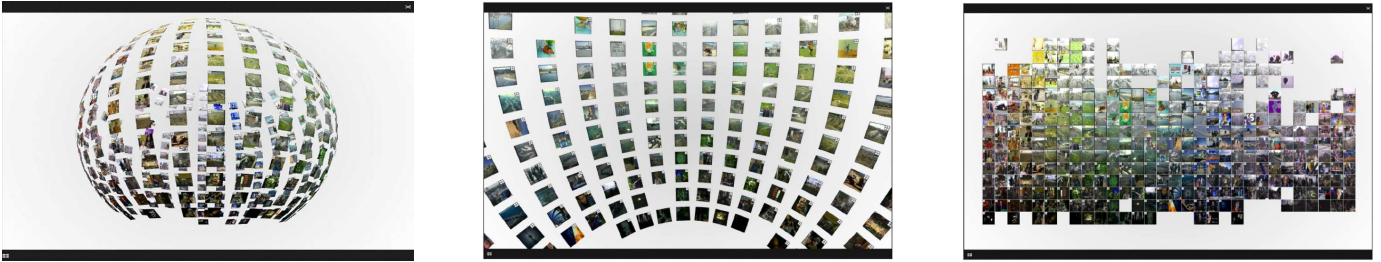


Fig. 1. Visualisation of MPEG-7 database projected onto hue sphere (left), with user immersed at the centre of the sphere (middle), and displayed on a flat plane (right).

resolution of the grid layout represents the maximal number of clusters present at a given level (which will only be met if all cells are filled). The grid cells then directly determine which images fall into which clusters, and assigning images to clusters/grid cells can hence be performed with simple comparison operations only. A representative image for each cluster is chosen by identifying the centroid image, that is the image for which the cumulative distance to all other images in the cluster is minimal.

This procedure is adopted at the each level of the tree hierarchy, i.e. first at the root node (the initial global display) and then for each non-empty cell again in a recursive manner, where the images of each child node are again mapped to a grid structure, until the complete tree is derived. The resulting interface provides an intuitive way of browsing so that the user can, starting from the initial display, successively select a representative image to refine the search. That image cluster is then expanded in the next level of the display where the user can again select an image group to navigate further into the image collection.

The grid-tree structure also provides another advantage. As the structure is fixed it can be pre-computed in completeness off-line, together with all possible grid view configurations the user can encounter. This in turn provides the user with the possibility of real-time browsing large image collections.

III. IMAGE BROWSING FOR ANNOTATION

In order to provide an effective tool for today's networked society, we have implemented an on-line version in Adobe Flash® CS 4 Professional [12]. Thus, it can be used on any machine with an Internet browser and Adobe Flash Player. Image data is stored in XML format, making it easily readable using appropriate tools and also providing the possibility of generating database structures using external software of choice.

In addition to the original hue sphere which provides an interaction scenario similar to a globe [9], [10], [11], we also implemented a second visualisation space whereby the user is placed at the centre of the sphere looking outwards thus being immersed in the visualisation itself. Furthermore, a flat visualisation space is also provided where images are projected onto a plane. Examples of these three paradigms, applied to the MPEG-7 common colour dataset [16], are given in Figure 1.

Navigation of the sphere gives the user the ability to rotate and tilt the sphere at the same time. Rotation around the y-axis is applied to the holder object, however tilting around the x-axis is applied to the sphere object therefore maintaining the direction of the applied tilt whilst rotating. Both rotation and tilting operations are smoothed via the introduction of an easing formula, recalculating and decelerating the rotational values every frame until they equal the newly assigned value. A limitation is applied to the tilting action to prevent the user viewing the sphere upside down.

The hierarchical data structure coupled with the corresponding organisation of images on a grid at various levels, allows for efficient and effective access to large databases. By selecting one image, the user can expand its image cluster, and in return the application will display the corresponding part of the next level of the hierarchy. While browsing through lower levels of the grid structure, the user can also immediately move to neighbouring image clusters. Where no immediate neighbour exists (i.e. when there are no images in the corresponding grid cell), the application will search adjacent cells and display an appropriate image for navigation.

Since automatic annotation systems [1] still have a long way to go before becoming sufficiently accurate and hence useful in practice, we see image browsing tools as an attractive means for efficiently annotating image datasets. The reason for this is that, as mentioned, similar images are clustered together, or at least close to each other in the visualisation, so that annotating these images simultaneously will take fewer interactions compared to going through an image list one-by-one. Consequently, we provide annotation facilities in our application that support exactly this idea. Users can select multiple images and then attach keyword information to the selected set. Fully utilising the hierarchical data structure, by selecting an image representing an image cluster, all images in that cluster are selected at the same time, allowing for an efficient mechanism of tagging whole image sets. Figure 2 shows an example of multiple images and image clusters selected for annotation, while Figure 3 gives an example of the actual annotation dialog.

IV. EXPERIMENTAL EVALUATION

Although the usefulness of the employed image browsing tools is evident, objective evaluation of such systems is difficult [17]. Nevertheless, we performed some user-based tests



Fig. 2. Selecting multiple images and image clusters for annotation (selected images have a green border).

where we compared the effectiveness of our image database navigation system with a common browsing tool, namely MS Windows Explorer®. The task given the users was to annotate a set of images. In particular, users had to find and annotate all images of a certain category (with each category consisting of between 3 and 18 images). The results of this task are listed in Table I. From there it is clear that our similarity-based browsing application is proven to provide a faster interface for image annotation. The flags category was annotated very quickly as all 3 images are very similar and therefore very close together. However the stained glass windows took somewhat longer using the browsing application due to larger fluctuations within hue and value, spreading them over the visualisation space. It should however be noted, that the relatively fast times for Windows Explorer here are based on the fact that the relevant images had similar filenames (or rather image numbers) and hence the reported results probably represent a best case scenario for linear image displays.

V. CONCLUSIONS

In this paper we have presented an image database visualisation and browsing system as an effective and efficient tool for annotating image repositories. Images are projected onto either a spherical or a planar visualisation space in a similarity-based fashion where visually similar images are located close to each other in the browsing interface. Large datasets are handled through a hierarchical data structure,

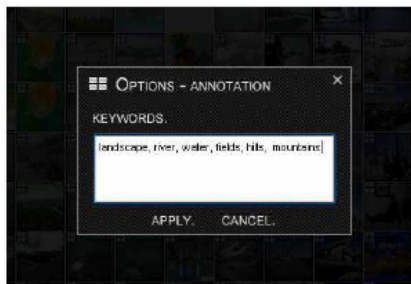


Fig. 3. Sample annotation dialog.

TABLE I
EXPERIMENTAL RESULTS OF ANNOTATION TASK.

		Windows Explorer		Browsing Application	
		Time Taken	No. Interactions	Time Taken	No. Interactions
1	USA Flags (3)	2:12	93	0:12	6
2	Stained Glass Windows (18)	5:13	258	3:18	41
3	Telephone Boxes (4)	2:42	121	1:01	25
Average		2:02	157	1:30	24

while images are arranged on a grid to avoid overlapping and hence user confusion. User-based experiments have shown that the browsing applications is more efficient for annotation than a system based on a conventional linear display of images.

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